([illegible]) آخر شابتر

Date : / / التاريخ : / /

Subject : Ch (5) Digital Filters موضوع الدرس :

FIR	IIR
Finite Impulse Response	Infinite Impulse Response,
$H(Z) = h(0) + h(1)Z^{-1} + h(2)Z^{-2}$ oR $= a_0 + a_1 Z^{-1} + a_2 Z^{-2}$ 2nd order	$H(Z) = \frac{a_0 + a_1 Z^{-1} + a_2 Z^{-1}}{1 + b_1 Z^{-1} + b_2 Z^{-2}}$
	- There is poles
- Found Realization structure (block Diagram) with direct method + Lattic	- Found Relazation structure (block diagram) with direct method only.

⟹ For FIR:

- Difference equation From Transfer function:

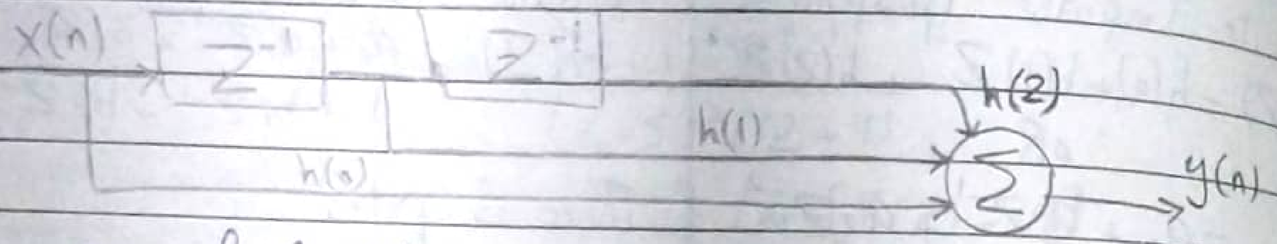
$$H(Z) = \frac{y(Z)}{X(Z)} = h(0) + h(1)Z^{-1} + h(2)Z^{-2}$$

$$y(Z) = h(0)X(Z) + h(1)X(Z)Z^{-1} + h(2)X(2)Z^{-2}$$

⟶ Difference eq.:

$$y(n) = h(0)X(n) + h(1)X(n-1) + h(2)X(n-2)$$

→ block Diagram (Direct method) Transuersal

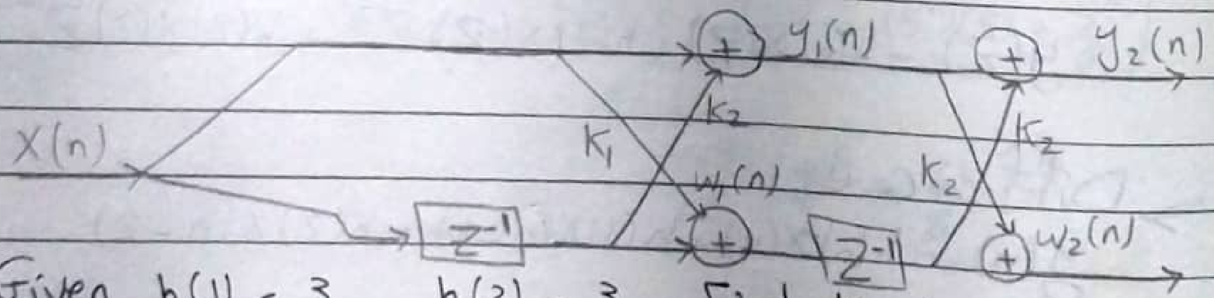


– No of Addition = Input to Summer − 1
$= 3 - 1 = 2$

– No of Mull = 3

– storage = I/P + Delay + Coeff
$2 + 2 + 3 = 7$

→ Lattice Methods.



Given $h(1) = 3$, $h(2) = 3$ Find K_1, K_2

From lattice: $y_1(n) = X(n) + K_1 X(n-1)$ → ①

$w_1(n) = K_1 X(n) + X(n-1)$ → ②

$y_2(n) = y_1(n) + K_2 w_1(n-1)$ → ③

From ②: $w_1(n-1) = K_1 X(n-1) + X(n-2)$ → ④

From (3,4) ⇒ $y_2(n) = X(n) + K_1 X(n-1) + K_1 K_2 X(n-1) + K_2 X(n-$

$y_2(n) = X(n) + (K_1 + K_1 K_2)(X(n-1)) + K_2 X(n-2)$ → I

Difference eq for FIR:

$y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2)$ II

Comparing: I & II

$\Rightarrow h(1) = K_1 + K_1 K_2$

$h(2) = K_2 = 3$　　$\boxed{K_2 = 3}$

$\rightarrow h(1) = 3 = K_1 + 3K_2$

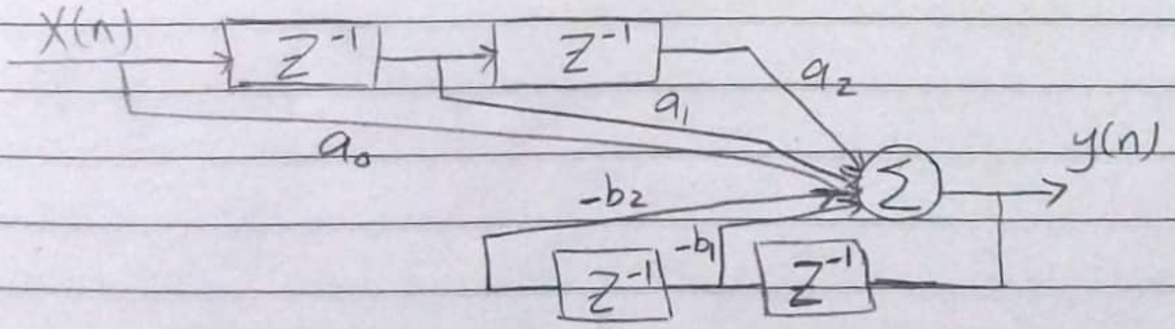
$\Rightarrow 3 = K_1 + 3 * 3$　　$\boxed{K_1 = -6}$

$\Rightarrow$ IIR Transversal Method (Direct method)

$$H(Z) = \frac{a_0 + a_1 Z^{-1} + a_2 Z^{-2}}{1 + b_1 Z^{-1} + b_2 Z^{-2}} = \frac{y(Z)}{x(Z)}$$

$\rightarrow y(Z)[1 + b_1 Z^{-1} + b_2 Z^{-2}] = X(Z)[a_0 + a_1 Z^{-1} + a_2 Z^{2}$

$y(n) = a_0 x(n) + a_1 x(n-1) + a_2 x(n-2) - b_1 y(n-1) - b_2 y(n-2)$



No of Addition = 4　　No of Mult = 5

storage = 2 + 4 + 5 = 11

Compare between FIR & IIR:

View of Comparison	FIR	IIR
No.of Mult = No of Coeff	3 h(0) -- h(2)	5 $a_0 \rightarrow a_2$ $b_1 \rightarrow b_2$
No of Addition – I/p to Sum -1	3 - 1 = 2	5 - 1 = 4
Storage Memory	3 Coeff + 2 shift register + 2 I/p - 7	5 Coeff + 2 shift + 2 I/p - 9